INCH-POUND

MIL-C-70461D (AR)
AMENDMENT 2
09 June 1993
SUPERSEDING
AMENDMENT 1
13 May 1992

# MILITARY SPECIFICATION

## COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS FOR CHARGE, PROPELLING, 155MM, M203A1

This Amendment forms a part of Military Specification MIL-C-70461D (AR) dated 28 May 1991, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

PAGE 4

* 3.4 TABLE I: Delete all the entries under the heading of tensile strength and substitute with the following new entries:

| | "Tensile Strength |
|---|---|
| Rear Cap | 1614 psi (Minimum) |
| Igniter Container | 812 lbs. (Minimum Peak Load) |
| Body, Molded | 2600 psi (Minimum)" |

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* 4.4.2.1 Major 108: Substitute "4.5.4" with "4.5.4.1".

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* 4.4.2.2 Add the following:

"Major 118 Tensile Strength 4.4.3.6 3.4 4.5.4.2"

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* 4.4.2.3 Add the following:

"Major 114 Tensile Strength 4.4.3.6 3.4 4.5:4.3"



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* 4.4.3.6 Delete in its entirety and substitute the following new paragraph:

"4.4.3.6 Tensile strength. Five (5) bodies, five (5) igniter containers and five (5) rear caps shall be randomly selected from the first article and each product lot of its respective component to provide the required test specimens for tensile tests in accordance with 4.5.4. Failure of any specimen shall be cause for lot rejection of its respective component."

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4.5.3.3.2.2 Delete in its entirely and substitute new paragraph 4.5.3.3.2.2:

"4.5.3.3.2.2 Method 2 (UV method). Cut 1.0 to 1.1 grams of sample into 1/4 inch squares or smaller. Dry in microwave moisture analyzer. Transfer the dried sample to a weighing bottle. Cool the sample to room temperature in a desiccate and weigh. Transfer the sample to a square homogenizing beaker. Reweigh the empty weighing bottle. Sample weight = total weight minus weight of empty weighing bottle. Add 50 ml of 90% methanol to the square beaker. Agitate the sample/solvent mixture in the hot ultrasonic bath for 10 minutes. Homogenize the mixture using a Binkmann homogenizer (or equivalent) for 2 minutes or until the sample is completely defiberized. Transfer the mixture to a filtering apparatus fitted with a suitable filter paper.

Add fresh 90% methanol to the square beaker. Grind this solution in the polytron for 1 minute. Add to the filtering apparatus. Repeat two times. Transfer the filtrate into a 250 ml volumetric flask. Make up to the mark with 90% methanol. Shake the flask well enough to make sure the solution is uniform. Using a pipet, transfer exactly 15 ml of this solution into a 50 ml volumetric flask. Make up to mark with 90% methanol. Shake well. Measure the absorbance of this solution at 285 nm. Calculate the DPA content according to the following formulas:

$$\text{DPA \%} = \frac{(Y - a)\ 250}{b * w}$$

Where y = absorbance
a = intercept of calibration curve
b = slope of calibration curve
w = sample of weight in gms."



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Add the following new paragraph 4.5.3.3.2.2.1:

"4.5.3.3.2.2.1 Calibration procedure. Dry approximately 1 gram of diphenylamine using the microwave moisture analyzer.

Weight approximately 200 mg to the nearest 0.2 mg. Add this to a 250 ml volumetric flask. Add 90% methanol solvent and dilute to mark. Mix well.

Pipet 5 mls from the 250 ml flask and transfer to a 100 ml volumetric flask. Dilute to mark using the same solvent and mix well. This is the standard stock solution. Volumetrically pipet the following aliquot of the standard solution into 100 ml volumetric flasks.

5 ml solution in a 100 ml volumetric flask = 0.002 mg/ml
10 ml solution in a 100 ml volumetric flask = 0.004 mg/ml
15 ml solution in a 100 ml volumetric flask = 0.006 mg/ml

Dilute all standard solutions to mark and mix well. Zero the spectrophotometer with fresh 90% methanol solvent at 285 nm. Rinse the cell with the first standard twice, then measure the absorbance.

Rinse the same cell with the second standard, then measure its absorbance. Perform the same procedure with the 3rd standard. Input the absorbance and concentration data points into a linear regression program. Obtain the slope, y-intercept and r value (correlation coefficient). If r is less than 0.999, the standard should be rerun."

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* 4.5.4 Delete in its entirety and substitute the following new paragraphs:

"4.5.4 Tensile strength

4.5.4.1 Body. Two specimens shall be prepared from each body. The specimens shall be taken from the center of the wall of the body. Test specimens shall be taken 180 degrees apart. The specimens shall have a one-inch gage length and width, but wall thickness shall fall within the tolerance specified on the applicable drawing. The test specimens shall be preconditioned for a minimum of 4 hours at 23 plus or minus (±) 2 degrees centigrade (C) at a relative humidity of 50 ± 5 percent (%). The tensile breaking strength of the specimens



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shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute. If any specimen fails to meet the applicable requirement, the lot of the body shall be rejected.

4.5.4.2 Igniter container. The igniter container shall be weighed to the nearest 0.1 g. The igniter container shall be cut a 1.5 inch diameter hole in top area and cut several V-slots into the bottom of the igniter container sample. The sample will then be placed in a test fixture that can be mounted in a tensile testing machine with a constant rate of crosshead movement (See Figure A). The tensile breaking strength of the specimen shall be determined by utilizing a crosshead speed of 1-inch per minute. Record the peak load and weight of each specimen. If any specimen fails to meet the applicable requirement, the lot of the igniter container shall be rejected.

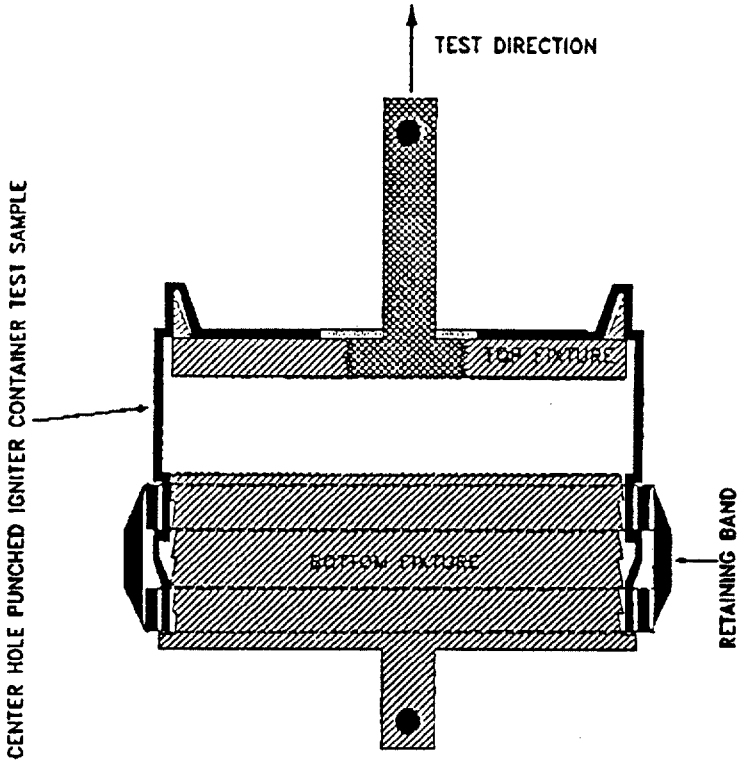



FIGURE A



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4.5.4.3 Rear cap. Two specimens shall be prepared from each rear cap. Test specimens shall be taken from the rear flat of the rear cap. Test specimens shall be taken 180-degree apart. The specimens shall have a 1-inch gage length and width and wall thickness shall fall within the tolerance specified on applicable drawing. Test specimens shall be preconditioned for a minimum of 4 hours at 23 degrees ± 2 degrees C and a relative humidity of 50 ± 5 %. The tensile testing machine shall be of a constant rate of crosshead movement type. Tensile strength shall be determined at a crosshead speed of 1-inch per minute. Record the tensile strength of the specimen. If any specimen fails to meet the applicable requirement, the lot of the rear cap shall be rejected."

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